Exercise sheet: Week 4

1. Recall Example 4 related to the asymptotic properties of MLEs. We had a random sample from Exp(λ) and data $n = 10$, $\bar{x} = 1.73$. We found that $\hat{\lambda} = 0.578$ and we used asymptotic properties to find an approximate 95% CI (0.357, 1.520) for λ.

   (a) Suppose now that the distribution is reparameterised in terms of $\theta = \frac{1}{\lambda}$, with p.d.f. $f_X(x;\theta) = \frac{1}{\theta}\exp(-x/\theta)$.

   Find the MLE $\hat{\theta}$ and use asymptotic properties to obtain an approximate 95% CI for θ. Evaluate the CI for the same data as above. Check that the MLEs agree (under the mapping $\theta = 1/\lambda$) but that the CI's do not agree.

   (b) For both parameterisations (i.e. λ and $\theta = 1/\lambda$), find the 95% CIs based on the asymptotic distribution of the deviance $D \sim \chi^2_1$. Show that the CIs do agree in this case. It should be clear that they do since both CIs are based on 'thresholding' the log-likelihood function.

2. (a) Suppose that $\underline{x}$ is a random sample of size n from a Poi(μ) distribution. Show that the deviance is given by
$$D(\mu) = 2n[(\mu - \bar{x}) + \bar{x}\log(\bar{x}/\mu)].$$

   (b) Hence, use R to find an approximate 99% CI for μ in the case where a sample $\underline{x}$ of 100 values gives $\sum x_i = 74$.

3. Suppose that we have a random sample of any finite size from the normal distribution N(μ, σ^2), where σ^2 is known.

   (a) Consider the asymptotic normality of the MLE and the further approximation replacing $I(\mu)$ by $I(\hat{\mu})$. Show that in this specific case both results hold *exactly* for all n and not just as $n \to \infty$.

   (b) Show that the asymptotic distribution of the deviance is also exact for this specific case, i.e. that $D(\mu) \sim \chi^2_1$ holds exactly for all n and not just as $n \to \infty$.

   [Hint: Recall the sampling properties of $\bar{X}$ when $\underline{X}$ is a sample from a normal distribution.]

4. Let $\underline{X}$ be a random sample of size n from the Poisson distribution with mean θ.

   Use the Central Limit Theorem to write down a large-sample approximate distribution for the MLE, and determine approximate 95% confidence intervals for θ by using the approximations

   (a) $\frac{\bar{X}-\theta}{\sqrt{\frac{\theta}{n}}} \approx N(0,1)$

   (b) using $\hat{\theta}$ in place of θ in the denominator on the l.h.s. of (a).

   A random sample of size 20 gives $\sum x_i = 94$. Calculate both confidence intervals and comment on their comparison.